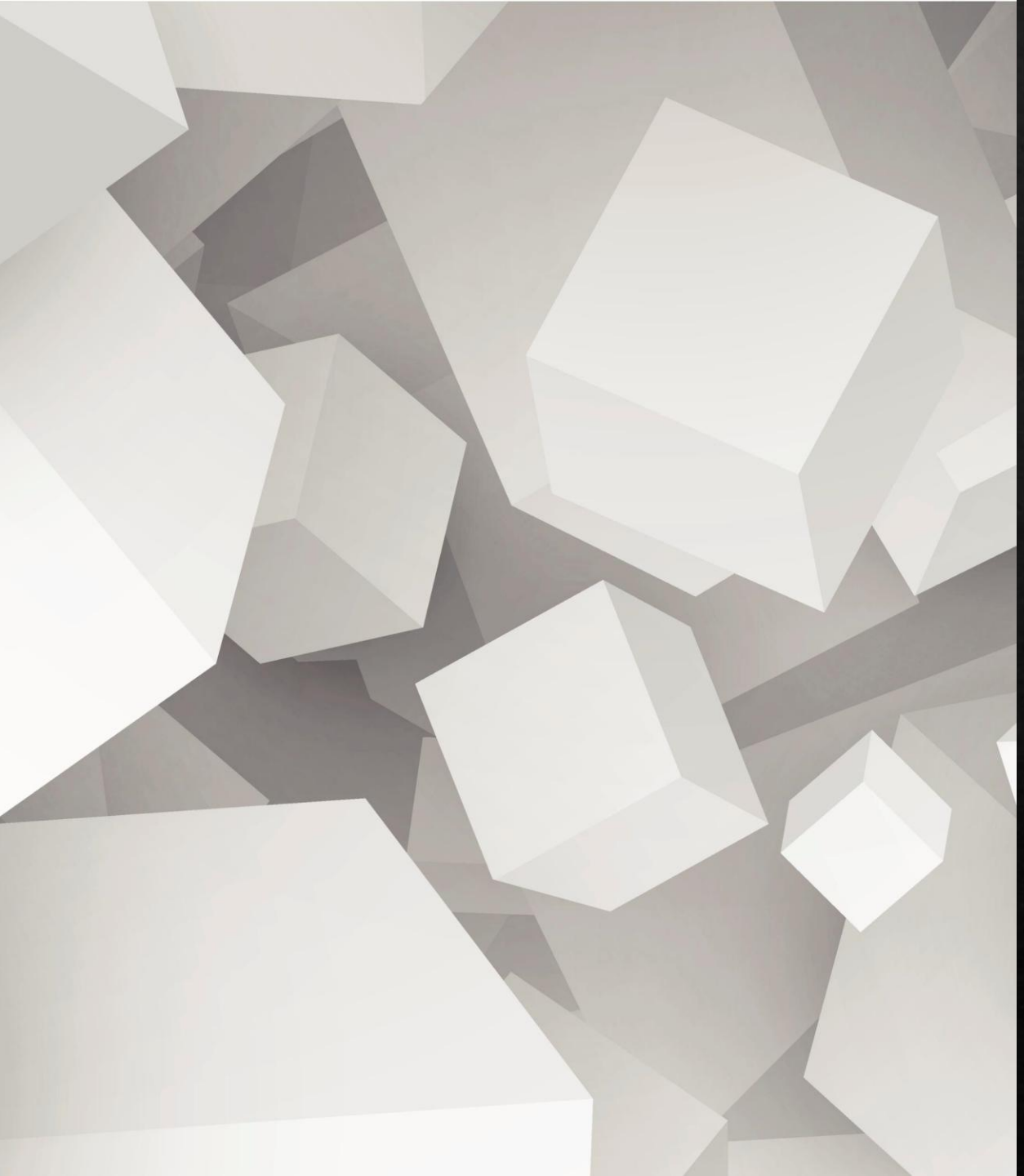


TASK 3: DATA EXPLORATION - CONCLUSIONS

TEAM NAME: Validated Encoders
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Biases

- **Device bias:** a small number of smartphone models dominate the dataset.

  Risk: classifiers may introduce device-specific frequency responses and noise floors instead of actual sounds.

- **Environment bias:** certain classes are strongly tied to specific rooms (e.g., *running_water* to kitchens).

  Risk: models might learn room acoustics rather than the target sound event.

- **Annotator bias:** With mostly two annotators per file, systematic errors directly impact labels.

Dataset Suitability

Verdict: Suitable for training Sound Event Detectors (SED), but with notable challenges.

Key Challenges:

- Severe Class Imbalance: e.g., *footsteps* appear in 9 times more in files than *coffee_machine*.
- Overlapping Profiles: Acoustically similar pairs (footsteps vs. keyboard typing) make pecise discrimination difficult.
- Scale Variance: Extreme differences in feature scales make normalisation mandatory.
- Dataset Strength: High annotator agreement (kappa=0.79) ensures generally reliable labels.

Broader Impact

Positive Applications: Enables beneficial technologies like elderly monitoring and smart home automation.

Risks of Misuse: Presents severe privacy risks, particularly unauthorised surveillance.

Mitigation Strategies:

- Ensure complete anonymity of data contributors.
- Limit dataset redistribution and usage.
- Enforce strict ethical guidelines and agreements.
- Explicitly prohibit surveillance purposes within the dataset license.